

A Appendix

A.1 Notation

Symbol	Description
<i>Indices, Sets, and Regions</i>	
i	Mobile-agent index, $i = 1, \dots, N$
t	Time index, $t = 0, \dots, H$
r	Region index, $r \in \{\text{top}, \text{bottom}\}$
b	Tax-bracket index, $b = 1, \dots, B$
N	Number of mobile agents
H	Episode horizon / simulation length
B	Number of tax brackets
$\mathcal{I}_{r,t}$	Set of agents located in region r at time t
$r(i, t)$	Region in which agent i is located at tax collection time
<i>State, Actions, and Policies</i>	
s_t	Full environment state at time t
$\mathbf{a}_{i,t}$	Multi-discrete action vector chosen by agent i at time t
$\boldsymbol{\tau}_{r,t}$	Tax-schedule vector chosen by planner r at time t
$\tau_{r,t}^b$	Marginal tax rate chosen by planner r for bracket b
π_{θ}	Shared mobile-agent policy
$\boldsymbol{\theta}$	Shared neural-network parameter vector for all mobile agents
π_{ϕ_r}	Policy of regional planner r
$\boldsymbol{\phi}_r$	Neural-network parameter vector of planner r
$\mathbf{h}_{i,t}$	Recurrent hidden-state vector of mobile agent i
$\mathbf{h}_{r,t}^P$	Recurrent hidden-state vector of planner r
$\mathbf{m}_{i,t}$	Agent action-mask vector at time t
$\mathbf{m}_{r,t}$	Planner action-mask vector at time t
γ	Discount factor
<i>Agent Observations</i>	
$\mathbf{o}_{i,t}^{\text{world}}$	Agent i 's local spatial grid observation
$\mathbf{o}_{i,t}^{\text{idx}}$	Agent i 's local spatial index map
$\mathbf{o}_{i,t}^{\text{agent}}$	Agent-specific non-spatial state vector
$\mathbf{o}_{i,t}^{\text{market}}$	Agent i 's market observation vector
$\mathbf{o}_{i,t}^{\text{tax}}$	Agent i 's tax-policy observation vector
<i>Planner Observations</i>	
$\mathbf{o}_{r,t}^{\text{world}}$	Regional spatial map observed by planner r
$\mathbf{o}_{r,t}^{\text{idx}}$	Regional index map observed by planner r
$\mathbf{o}_{r,t}^{\text{tax}}$	Planner r 's regional tax-state observation vector
$\mathbf{o}_{r,t}^{\text{market}}$	Planner r 's local market-state observation vector
$\mathbf{o}_{r,t}^{\text{agents}}$	Planner-side per-agent entries for agents in region r

Table 1: Notation for indices, policies, actions, and observations.

Symbol	Description
<i>Coin, Income, and Labor</i>	
$x_{i,t}^c$	Coin endowment of agent i at time t
$\mathbf{x}_t^c$	Vector of coin endowments across all agents at time t
$\mathbf{x}_{r,t}^c$	Coin-endowment vector for agents located in region r at time t
$x_{i,t}^{c,\text{inv}}$	Available inventory coin of agent i at time t
$\bar{x}_{i,t}^c$	Coin snapshot of agent i from the start of the tax period
$z_{i,t}$	Period income of agent i before truncation at zero
$z_{i,t}^+$	Taxable income of agent i , defined as $\max\{z_{i,t}, 0\}$
$l_{i,t}$	Cumulative labor of agent i at time t
<i>Utility, Welfare, and Rewards</i>	
$\text{CRRA}(x_{i,t}^c)$	CRRA utility from coin consumption
η	Coefficient of relative risk aversion
κ	Labor-disutility parameter
$u_i(x_{i,t}^c, l_{i,t})$	Utility of agent i at time t
$\rho_{i,t}$	Reward of agent i at time t
$\rho_{r,t}^P$	Reward of planner r at time t
$\text{gini}(\mathbf{x}_{r,t}^c)$	Gini coefficient for agents in region r
$\text{eq}_r(\mathbf{x}_{r,t}^c)$	Regional equality measure, $1 - \text{gini}(\mathbf{x}_{r,t}^c)$
$\text{prod}_r(\mathbf{x}_{r,t}^c)$	Regional production measure
$\text{swf}_{r,t}(\mathbf{x}_{r,t}^c)$	Regional social welfare function for planner r
<i>Taxation and Redistribution</i>	
m_b	Lower cutoff of tax bracket b
$\mathbf{m}$	Vector of bracket cutoffs, $\mathbf{m} = (m_1, \dots, m_{B+1})$
$T_r(z_{i,t}^+)$	Tax liability under region r 's tax schedule
$\hat{T}_{i,t}$	Effective tax payment made by agent i
$R_{r,t}^{\text{tax}}$	Regional income-tax revenue in region r
$R_{r,t}^{\text{travel}}$	Regional revenue from travel payments
$R_{r,t}^{\text{tariff}}$	Regional revenue from import tariffs
$R_{r,t}$	Total regional revenue in region r
$g_{r,t}$	Lump-sum transfer per agent in region r

Table 2: Notation for economic variables, welfare, rewards, and taxation.

A.2 Plots

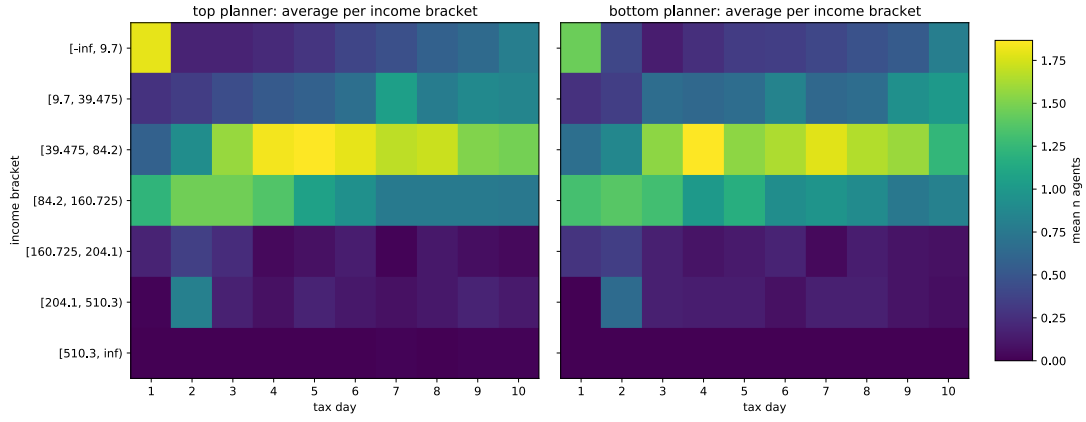
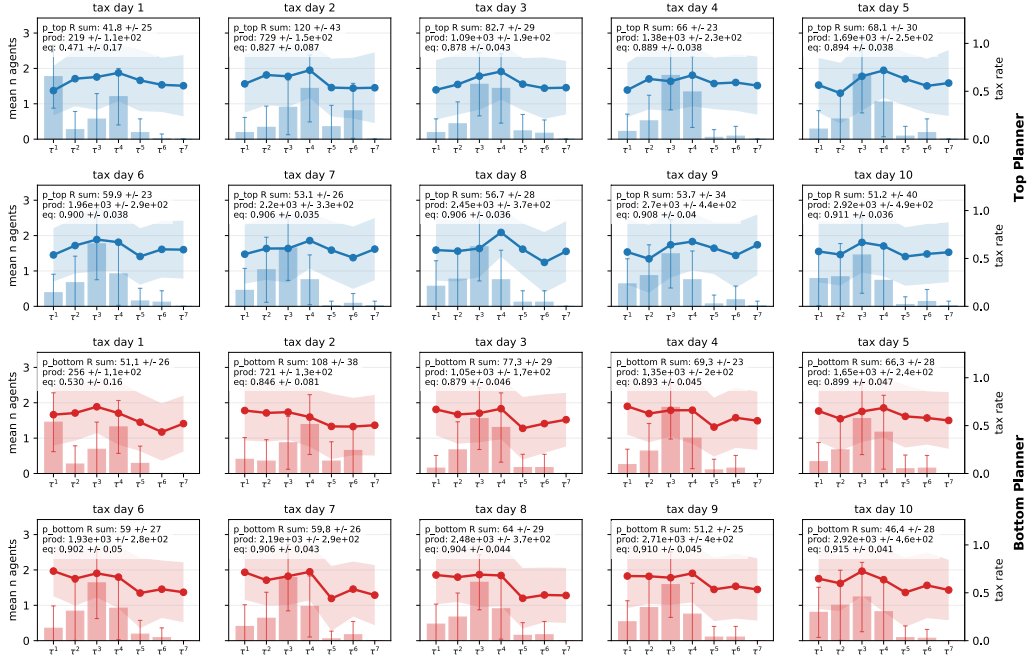
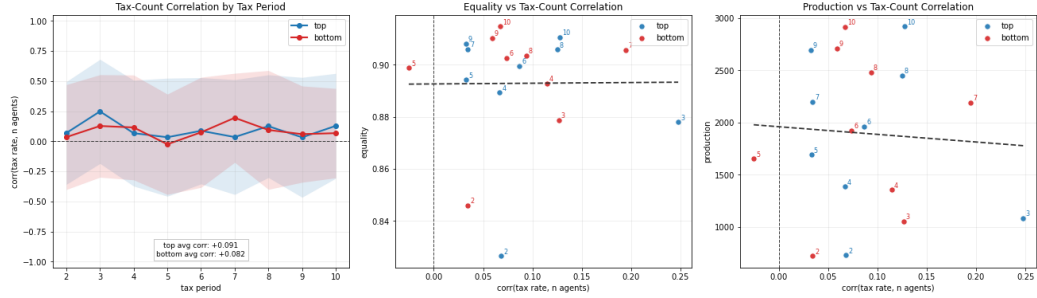


Figure 22: **Average income-bracket distribution in the baseline setting.** The heatmap shows the average distribution of agents across income brackets in the baseline experiment, averaged over rollout trajectories and training seeds. The figure provides a reference for interpreting planner tax schedules by showing which brackets are frequently populated and which brackets are rarely reached.



(a) **Average tax-day snapshots across dense logs and seeds.** Bars show the mean number of agents in each income bracket, while lines show the mean marginal tax rate selected for each bracket.



(b) **Tax-bracket alignment with equality and production.** The left panel shows the tax-count correlation across tax periods, while the middle and right panels compare this correlation with equality and production. The correlation is calculated over brackets τ_1 – τ_6 . The highest bracket is excluded since it almost never reached and might impact averages unnecessarily (as also shown in Appendix Figure 22). The first tax period is also excluded from the main figure because it reflects an early transient phase of the simulation. The corresponding plot including the first period is reported in Appendix Figure 37.

Figure 23: **Planner tax schedules and tax-count alignment.** Panel (a) compares the number of agents in each income bracket with the marginal tax rates chosen by each planner. Panel (b) summarizes this relationship using the correlation between bracket population and marginal tax rates, and compares the correlation with equality and production.

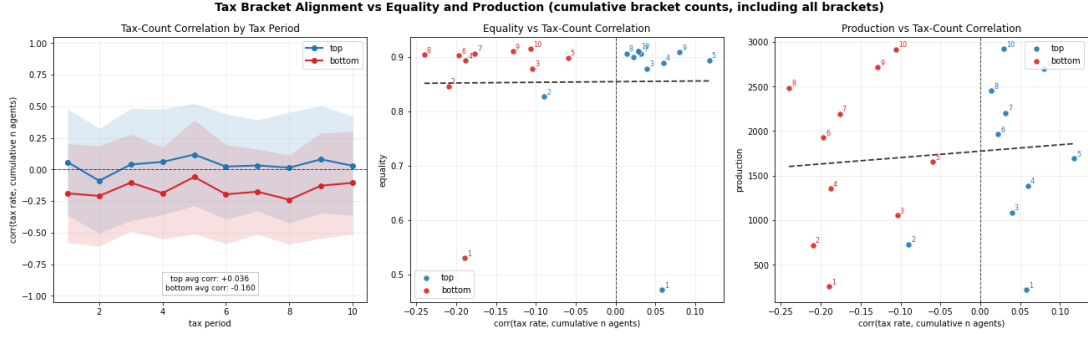


Figure 24: **Cumulative tax-count alignment in the baseline setting.** The figure shows the correlation between marginal tax rates and the cumulative number of agents reaching each income bracket. The cumulative measure is closer to the marginal tax mechanism, since it captures whether planners assign higher rates to brackets that agents actually reach. The systematic difference between planners appears in two out of three seeds and should therefore be interpreted as a descriptive training pattern rather than a robust planner asymmetry.

for bracket b , the cumulative count is

$$C_{r,t}^b = \sum_{j=1}^b n_{r,t}^j$$

where: $C_{r,t}^b$ is the cumulative number of agents up to bracket b . $n_{r,t}^j$ is the number of agents in bracket j . $r \in \{\text{top}, \text{bottom}\}$. t is the tax period. $b \in \{1, \dots, B\}$.

Giving the cumulative count vector:

$$C_{r,t} = (C_{r,t}^1, C_{r,t}^2, \dots, C_{r,t}^B)$$

and the correlation is:

$$\rho_{r,t} = \text{corr}(\tau_{r,t}, C_{r,t})$$

with:

$$\tau_{r,t} = (\tau_{r,t}^1, \tau_{r,t}^2, \dots, \tau_{r,t}^B)$$

The cumulative version is closer to the actual marginal tax mechanism: it asks whether the planner assigns higher marginal rates to brackets that agents actually reach.



Figure 25: **Market activity with and without cross-regional trade.** The figure compares market size, average transaction prices, and trade activity across settings with and without cross-regional trade. Market size is measured as the mean quantity of wood and stone held in escrow across runs. Large price movements in the earliest periods should be interpreted cautiously, since a small number of early market entries can strongly affect the average price when market activity is still limited.

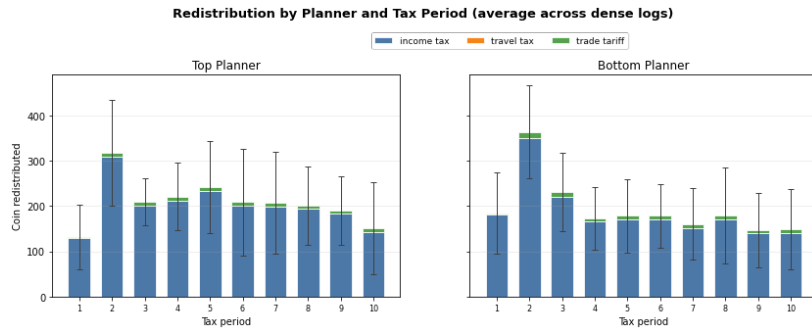


Figure 26: **Redistribution revenue in the trade-integrated setting.** The figure decomposes regional redistribution into income-tax revenue and import-tax revenue in the cross-regional trade experiment. The plot shows how market integration introduces an additional revenue channel through import taxation, although income taxation remains the main source of redistribution in most periods.

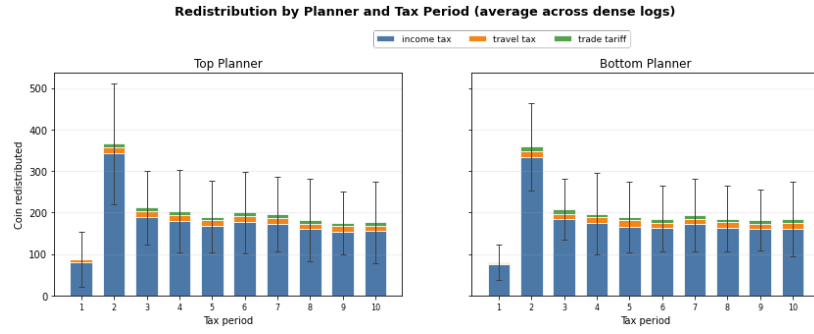


Figure 27: **Redistribution revenue in the travel enabled setting.** The figure decomposes regional redistribution into income-tax revenue, import-tax revenue, and travel revenue in the travel-enabled experiment. The plot shows how mobility adds a further revenue channel, while also changing the regional population over which redistribution is allocated.

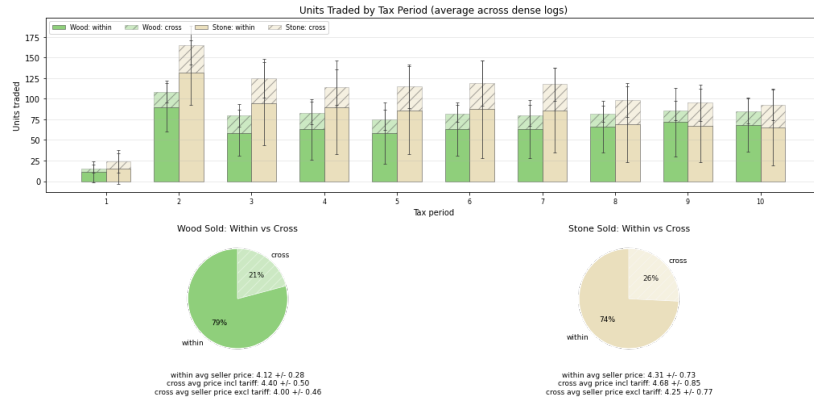


Figure 28: **Trade composition in the mobility setting.** The figure reports trade metrics for the travel-enabled experiment (comparable to the trade enabled setting shown in figure 12). Compared with the trade-integrated setting, stone accounts for a larger share of trading activity, although the standard-deviation bands indicate substantial variation across runs. This suggests that mobility changes trade composition, but that the pattern is not uniform across training seeds.

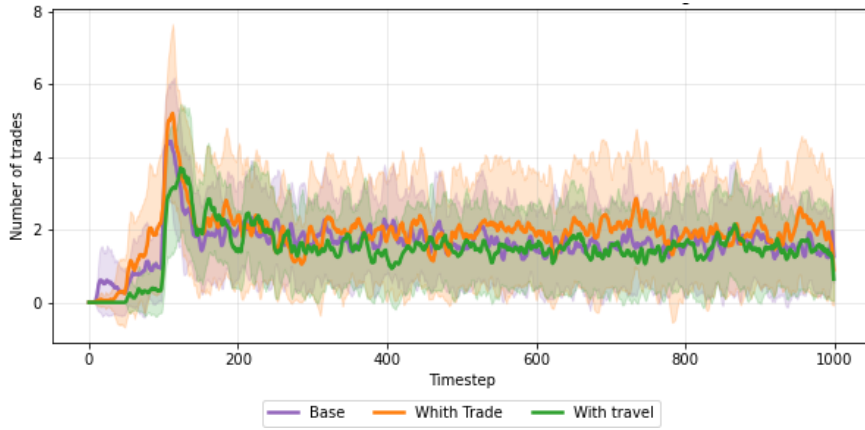


Figure 29: **Trade activity across experimental settings.** The figure compares trade counts over time in the baseline, trade-integrated, and mobility settings. Lines show mean trade counts across dense logs, while shaded areas indicate one standard deviation. The comparison shows how cross-regional trade and mobility change the level and variability of market activity relative to the closed-economy baseline.

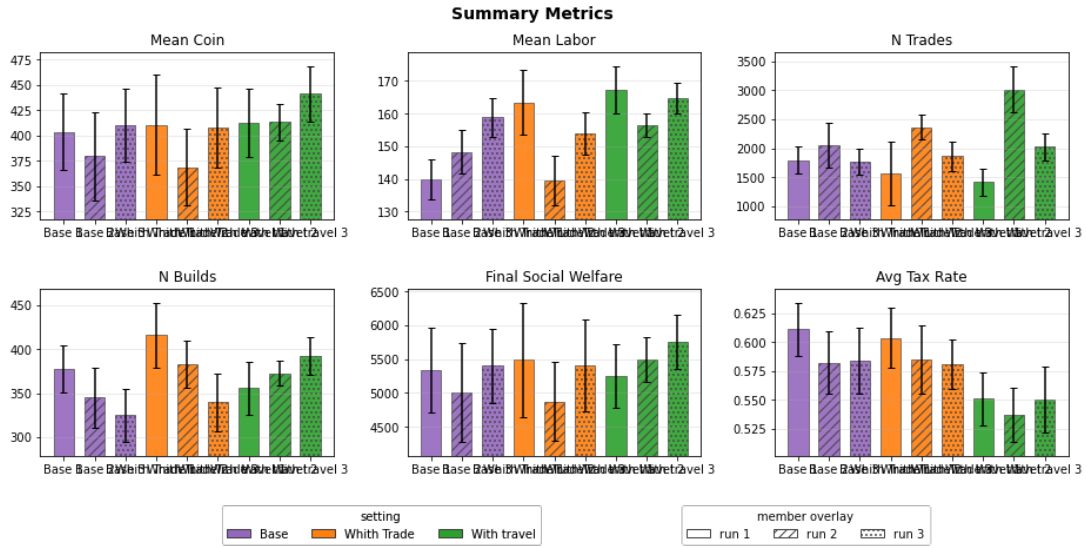


Figure 30: **Summary metrics across training seeds.** The figure reports key outcome metrics separately for each simulated training seed across the experimental settings. Showing the seed-level outcomes highlights the extent of variation across independently trained policies and supports the interpretation of the results as descriptive patterns rather than statistically definitive estimates.

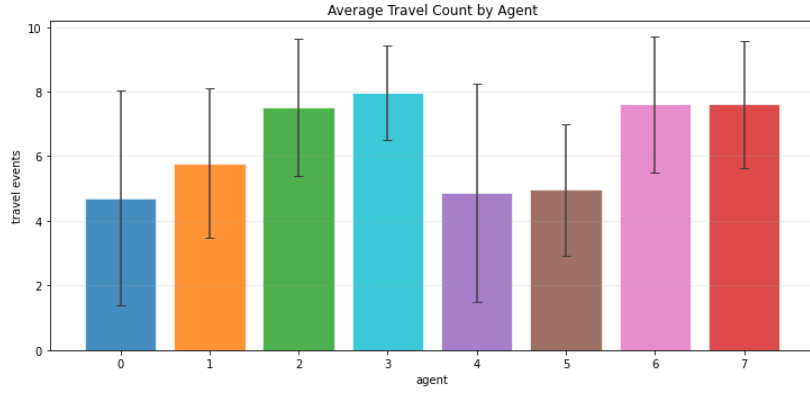


Figure 31: **Relocation events by agent.** The figure shows the average number of relocation events by agent index across the three training seeds. The distribution illustrates that mobility is not used uniformly across agents, suggesting that relocation behavior depends on agent-specific conditions such as skill, location, accumulated resources, and learned policy behavior.

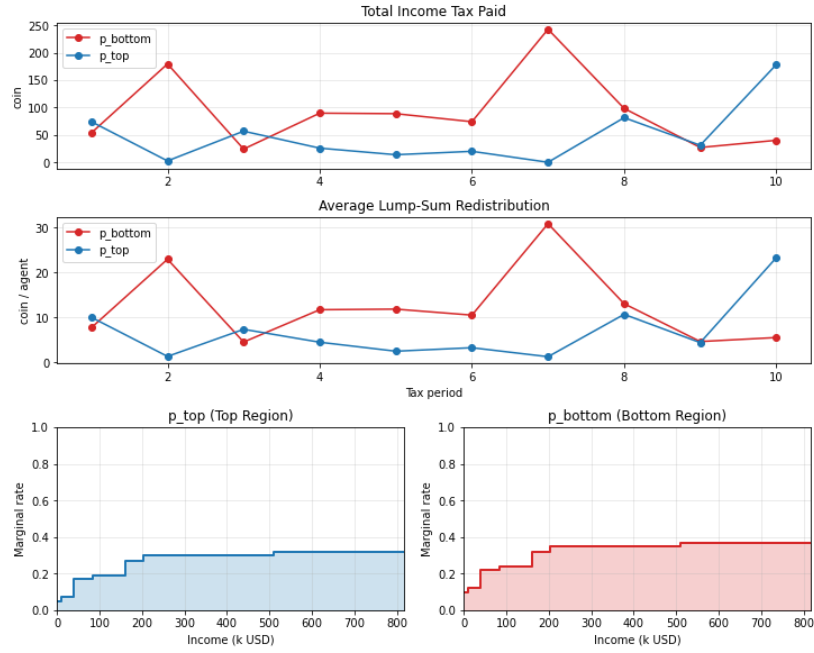


Figure 32: **Fixed-tax experiment outcomes.** The figure shows income and redistribution across tax periods and regions in the fixed-tax experiment. The lower panel reports the fixed tax schedules used in the experiment, where the top region has tax rates five percentage points lower than the bottom region in each bracket. The results provide a controlled comparison for examining whether a persistent tax advantage is sufficient to generate systematic relocation toward the lower-tax region.

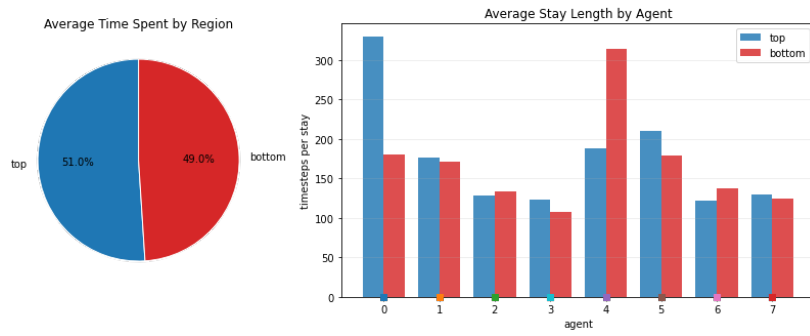


Figure 33: **Residence patterns in the regular mobility setting.** The figure reports the same residence measure as Figure 19, but for the regular travel-enabled setting without fixed tax schedules. The comparison provides a baseline for interpreting the fixed-tax experiment by showing how agents distribute across regions when tax schedules are learned rather than externally imposed.

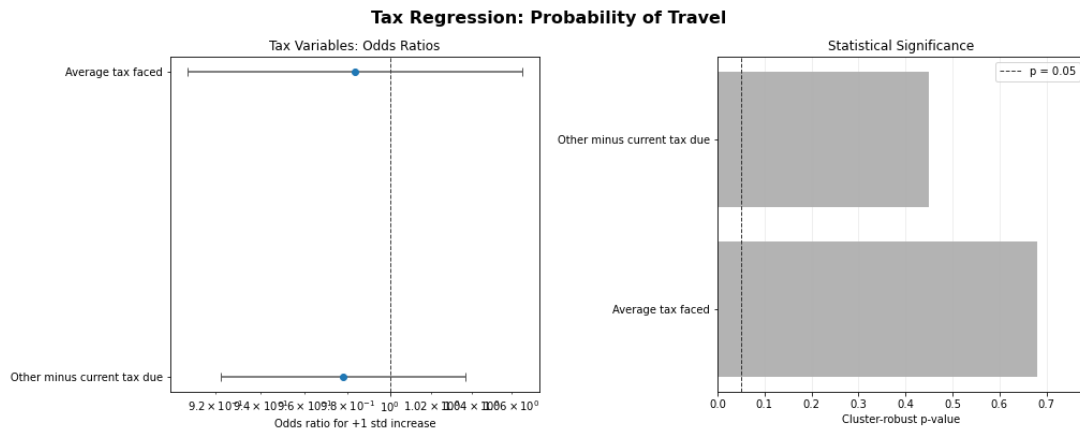


Figure 34: **Estimated tax effects in the relocation regression.** The figure reports odds ratios and p-values for the tax-related variables in the relocation regression. The estimates summarize whether current-region and alternative-region tax exposure are associated with relocation decisions, while the uncertainty measures indicate how cautiously these effects should be interpreted.

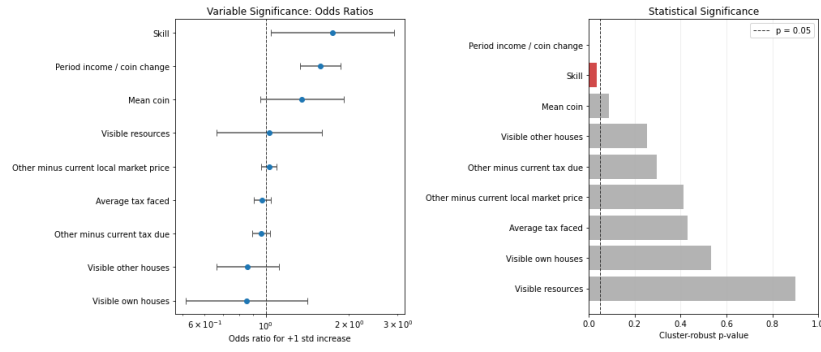


Figure 35: **Relocation regression without agent fixed effects.** The figure reports odds ratios and p-values from the pooled relocation regression, including tax-related variables but excluding agent fixed effects. The estimates are broadly consistent with the main regression results found in 17, suggesting that the relationship between relocation and observed economic conditions is not driven only by the inclusion of agent fixed effects.

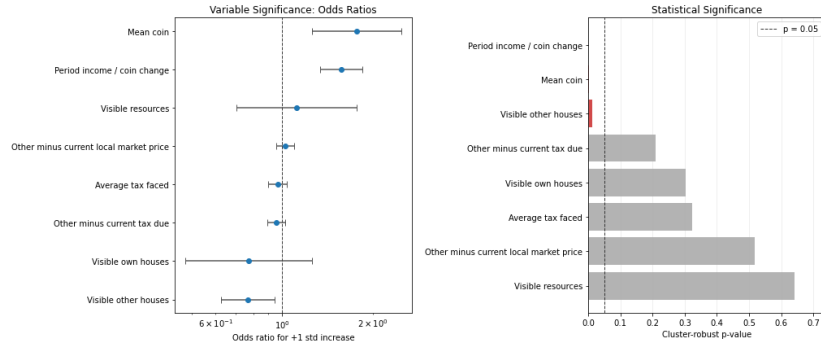


Figure 36: **Relocation regression with agent fixed effects.** The figure reports odds ratios and p-values from the relocation regression including agent fixed effects. This specification controls for persistent differences across agents, allowing the estimates to focus more directly on within-agent variation in tax exposure, income, resources, and spatial conditions. The results are consistent with the results found in 17.

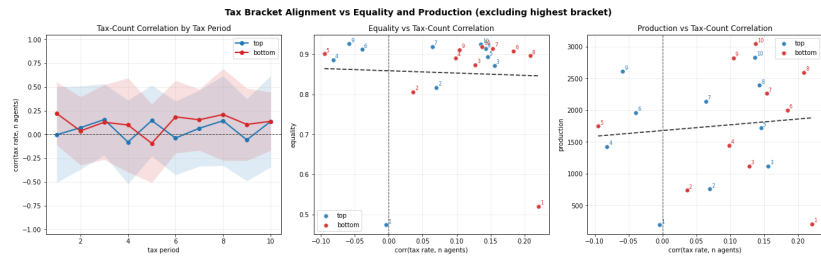


Figure 37: **Tax-count alignment including the first tax period.** The figure corresponds to the tax-count alignment analysis in panel (b) of figure 23, but includes the first tax period. The first period is visibly different from later periods, illustrating why it is treated as an early transient phase and excluded from the main tax-response analysis.

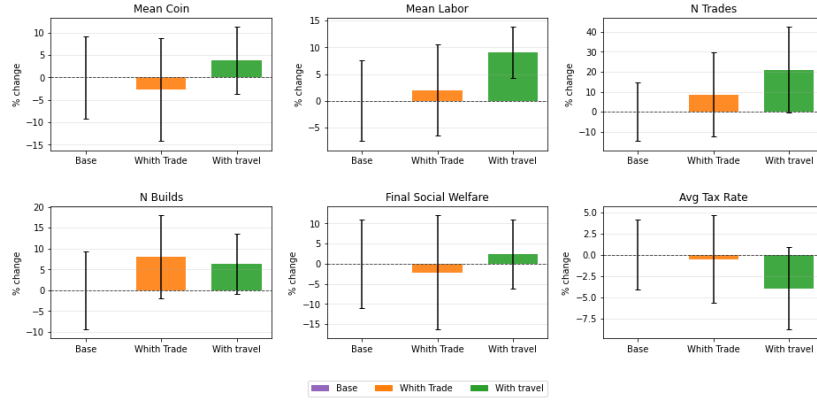


Figure 38: **Percentage differences in summary metrics.** The figure shows percentage differences in key summary metrics between the baseline and the trade-integrated and mobility settings. The comparison provides a normalized view of how the experimental extensions change aggregate outcomes relative to the closed-economy baseline.

A.3 Algorithm

Algorithm 1 summarizes the training procedure used in the simulations. The implementation follows RLLib’s multi-agent PPO setup. At each timestep, the shared mobile-agent policy chooses actions for all mobile agents, while each regional planner policy chooses a tax-action vector. Planner actions are sampled every timestep, but action masks restrict tax-rate changes to valid tax-decision times. The environment then advances one step, updates the economic state, and internally applies tax collection and redistribution whenever the tax-period condition is met.

Algorithm 1 Phased PPO training loop for mobile agents and regional planners

Input:

L (PPO rollout length),
 T (tax-period length),
 K (number of regions; $K = 2$ in the implemented simulations),
 $\mathcal{A}$ (PPO update rule),
 C (stopping criterion),
 $\mathcal{P}$ (training phase)

Output:

θ (shared mobile-agent policy parameters),
 $\{\phi_r\}_{r=1}^K$ (regional planner policy parameters)
Initialize environment state s_0 , agent observations $\{\mathbf{o}_{i,0}\}_{i=1}^N$, planner observations $\{\mathbf{o}_{r,0}^P\}_{r=1}^K$
Initialize recurrent states $\{\mathbf{h}_{i,0}\}_{i=1}^N$ and $\{\mathbf{h}_{r,0}^P\}_{r=1}^K$
Initialize policy parameters θ and $\{\phi_r\}_{r=1}^K$
Initialize rollout buffers $D^A \leftarrow \emptyset$ and $\{D_r^P\}_{r=1}^K \leftarrow \{\emptyset\}_{r=1}^K$
while training **do**
 for $t = 0, \dots, L - 1$ **do**
 for $i = 1, \dots, N$ **do**
 $\mathbf{a}_{i,t}, \mathbf{h}_{i,t+1} \sim \pi_\theta(\cdot \mid \mathbf{o}_{i,t}, \mathbf{h}_{i,t}, \mathbf{m}_{i,t})$
 end for
 for $r = 1, \dots, K$ **do**
 $\tau_{r,t}, \mathbf{h}_{r,t+1}^P \sim \pi_{\phi_r}(\cdot \mid \mathbf{o}_{r,t}^P, \mathbf{h}_{r,t}^P, \mathbf{m}_{r,t})$
 end for
 $s_{t+1}, \{\mathbf{o}_{i,t+1}\}_{i=1}^N, \{\mathbf{o}_{r,t+1}^P\}_{r=1}^K, \{\rho_{i,t}\}_{i=1}^N, \{\rho_{r,t}^P\}_{r=1}^K, d_t \leftarrow$
Env.step($s_t, \{\mathbf{a}_{i,t}\}_{i=1}^N, \{\tau_{r,t}\}_{r=1}^K$)
 Store agent transitions in D^A
 Store planner transitions in $\{D_r^P\}_{r=1}^K$
 Set s_t , observations, and recurrent states to their next values
 if d_t **then**
 Reset completed environment and recurrent states
 end if
 end for
 Estimate returns and advantages from D^A and $\{D_r^P\}_{r=1}^K$ using GAE
 if $\mathcal{P}$ trains mobile agents **then**
 Update θ using D^A and PPO update rule $\mathcal{A}$
 end if
 if $\mathcal{P}$ trains planners **then**
 for $r = 1, \dots, K$ **do**
 Update ϕ_r using D_r^P and PPO update rule $\mathcal{A}$
 end for
 end if
 Reset rollout buffers $D^A \leftarrow \emptyset$ and $\{D_r^P\}_{r=1}^K \leftarrow \{\emptyset\}_{r=1}^K$
 if criterion C is met **then**
 return $\theta, \{\phi_r\}_{r=1}^K$
 end if
end while

The phased PPO training loop alternates between rollout collection and policy optimization. During rollout collection, mobile agents and regional planners sample actions from their current policies, interact with the economy, and store the resulting observations, actions, rewards, value estimates, and recurrent states. After each rollout batch, returns and advantages are estimated using GAE. Depending on the active training phase, PPO updates the shared mobile-agent policy, the regional planner policies, or both. This structure allows agents and planners to co-adapt while also permitting selected phases in which one level of the system is held fixed.

In the implemented simulations, $K = 2$, corresponding to the top and bottom regions of the map. The eight mobile agents share the same policy parameters θ , while the two regional planners have separate policy parameters, ϕ_{top} and ϕ_{bottom} . Thus, experience from all mobile agents contributes to the update of the shared mobile-agent policy, whereas each planner is updated using its own planner-specific rollout data.

The notation $\mathbf{o}_{i,t}$ denotes the full observation available to mobile agent i , including its spatial map observation, index map, agent-specific state variables, market information, and tax information. Similarly, $\mathbf{o}_{r,t}^P$ denotes the full observation available to planner r , including its regional spatial observation, regional index map, tax state, market information, and planner-side per-agent entries. The masks $\mathbf{m}_{i,t}$ and $\mathbf{m}_{r,t}$ restrict the set of valid actions. For planners, these masks are especially important because they allow the planner policies to sample actions every timestep while only permitting tax-bracket updates at valid tax-decision times. At other timesteps, the valid planner action is effectively a no-op.

Tax collection and redistribution are not implemented as a separate call in the training loop. Instead, they are handled internally by `Env.step` through the regional tax components whenever the tax-period condition is met. Therefore, the rewards $\rho_{i,t}$ and $\rho_{r,t}^P$ already reflect the economic consequences of production, trade, travel, taxation, redistribution, and changes in utility or regional welfare.

The rollout buffers are written compactly in Algorithm 1. In practice, PPO stores additional quantities required for policy-gradient learning, including sampled actions, rewards, value-function predictions, action log-probabilities, done flags, action masks, and recurrent-state or sequence information. After each rollout batch is collected, returns and advantages are estimated using generalized advantage estimation. PPO then updates policy parameters by increasing the probability of actions with positive estimated advantage and decreasing the probability of actions with negative estimated advantage, while clipping the update to prevent the new policy from moving too far from the policy that generated the rollout data.

Which policies are updated depends on the training phase. Phase 1 trains the mobile-agent policy only. Phase 2 trains mobile agents and planners jointly. Phase 3A trains mobile agents while planner policy weights are held fixed; the planners still act in the environment, but their neural-network parameters are not updated. Phase 3B again trains mobile agents and planners jointly.

All trainable policies use the custom `keras_conv_lstm` architecture. The shared mobile-agent policy uses two convolutional layers, two fully connected layers with 128 hidden units, and an LSTM with 128 units. Each planner policy uses the same architecture but with two fully connected layers of 256 hidden units and an LSTM with 256 units. The actor and critic are implemented as separate policy and value branches.

A.4 Parameters

The key training hyperparameters used throughout the experiments are summarized in Table 3.

Table 3: Training hyperparameters for `simulation.run.py`

Parameter	Value
Number of rollout workers	4
Environments per worker	1
Parallel environment replicas	4
Sampling horizon per environment	200
Training batch size	800
SGD minibatch size	200
Number of SGD passes per batch	4
Minibatch gradient updates per iteration	16
CPUs per worker	1
GPUs	1
Framework	TensorFlow
Learning rate (agent)	0.0003
Learning rate (planner)	0.0001
Entropy regularization coefficient (planner)	0.02
Entropy regularization coefficient (agent, Phase 3A)	0.01
PPO clipping parameter ϵ	0.3
Episode length	1000
Tax period	100
SGD sequence length	25
Agent policy model	<code>keras_conv_lstm</code>
Planner policy model	<code>keras_conv_lstm</code>
Agent spatial input	11×11 local map
Planner spatial input	25×25 regional map
Agent convolutional layers	2 layers: 16, 32 filters
Planner convolutional layers	2 layers: 16, 32 filters
Convolutional kernel and stride	3×3 , stride 2
Index-map embedding dimension	4
Agent fully connected layers	2 layers, 128 hidden units
Planner fully connected layers	2 layers, 256 hidden units
Agent LSTM size	128 units
Planner LSTM size	256 units
Actor and critic structure	Separate policy and value branches
Agent action logits	57
Planner action logits	308 per planner
Phase 1 training duration	500 iterations, approx. 0.4M steps
Phase 2 training duration	1500 iterations, approx. 1.2M steps
Phase 3A training duration	500 iterations, approx. 0.4M steps
Phase 3B training duration	1500 iterations, approx. 1.2M steps
Phase 2 tax annealing schedule	(0, 0.0125)
Phase 2 tax annealing duration	400 iterations, approx. 320k aggregate steps
Travel enabled	No, in current <code>simulation.run.py</code> setup
Regional trade restriction	Yes
Fixed planner taxes	No

Table 4: Environment hyperparameters

Parameter	Value
World width	51
World height	25
Episode length	1000
Number of agents	8
Starting agent coin	0
Tax period	100
Planner subclasses	2
Build labor cost	2.1
Auction max bid/ask	10
Auction order labor	0.05
Auction max number of orders	10
Auction order duration	50
Gather move labor	0.21
Gather collect labor	0.21
Tax annealing schedule	[-100, 0.0125]

A.5 Tax table

Bracket	Income range	Planner action
1	$0 \leq z < 9.7$	τ^1
2	$9.7 \leq z < 39.475$	τ^2
3	$39.475 \leq z < 84.2$	τ^3
4	$84.2 \leq z < 160.725$	τ^4
5	$160.725 \leq z < 204.1$	τ^5
6	$204.1 \leq z < 510.3$	τ^6
7	$z \geq 510.3$	τ^7

Table 5: Income tax brackets corresponding to the US tax bracket system and used for most of the experiments. The bracket cutoffs are given by $m = [0, 9.7, 39.475, 84.2, 160.725, 204.1, 510.3, \infty]$.

A.6 Aneling Procedure

Training is organized into four phases. Phase 1 trains the mobile-agent policy while taxes and travel are disabled. This allows the agents to learn the basic gather, trade, and build dynamics before being exposed to regional mobility. Phase 2 trains mobile agents and planners jointly with regional taxation enabled. Phase 3A enables travel and trains the mobile agents while planner policy weights are held fixed. Phase 3B then trains both mobile agents and planners jointly with travel enabled.

The simulations use PPO with rollout fragments of length 200 and an episode length of 1000 timesteps. Thus, each rollout environment completes one full episode after approximately

$$\frac{1000}{200} = 5$$

PPO iterations. This conversion is useful for interpreting schedules that are defined in terms of completed episodes rather than PPO training iterations.

During Phase 2, the maximum tax rate available to the planners is gradually annealed. The implemented tax annealing schedule is

$$(0, 0.0125),$$

where the first element is the warmup period and the second element is the slope of the annealing schedule. The allowed maximum tax rate is computed as

$$\bar{\tau}_{\max} = 0.0125 \cdot (\text{completions} - 0),$$

bounded above by 1.0. Since the warmup period is zero, the tax cap begins increasing immediately. The cap reaches 100% when

$$1.0 = 0.0125 \cdot \text{completions}.$$

Solving for completions gives

$$\text{completions} = \frac{1.0}{0.0125} = 80.$$

Thus, the tax cap reaches 100% after approximately 80 completed episodes per rollout environment. Since one completed episode corresponds to approximately 5 PPO iterations, this is equivalent to

$$80 \times 5 = 400$$

PPO iterations. Therefore, under the implemented settings, the Phase 2 tax annealing schedule reaches the full tax range around Phase 2 iteration 400.

After the annealing period, planners can select from the full discrete tax-rate set. Before this point, action masking restricts the highest available tax rates, so planners gradually gain access to larger marginal tax rates over the course of training. This reduces the abruptness of early planner interventions and allows agents and planners to adapt gradually to the tax-policy action space.

A.6.1 Additional Implementation Specifications

The `keras_conv_lstm` model follows the architecture used in the AI Economist framework [12]. It combines convolutional layers for spatial observations, fully connected layers for the combined feature representation, and an LSTM for temporal information. The recurrent component allows each policy to maintain a hidden state over time, which is important in the partially observed multi-agent economy.

Dense logs saved after Phases 1, 2, and 3A are generated with deterministic action selection (`explore=False`), while the final Phase 3B dense logs used for analysis are generated with `explore=True`. The final reported rollouts therefore include stochastic policy sampling rather than purely deterministic greedy evaluation.

In the implemented configuration, planner policies use an entropy coefficient of 0.02. The shared agent policy uses RLlib’s default entropy coefficient of 0.0, except in Phase 3A, where the agent entropy coefficient is set to 0.01. This means that entropy regularization is applied asymmetrically across policies. Planner policies use a positive entropy coefficient to encourage exploration over the large discrete tax-policy space and reduce the risk of premature convergence to a narrow set of tax schedules. The shared agent policy uses no additional entropy bonus in most phases, relying instead on PPO’s stochastic action sampling. This avoids excessively noisy behavior in the agents’ already large multi-discrete action space. In Phase 3A, an agent entropy coefficient is introduced to encourage exploration of behavioral responses to fixed planner policies, including relocation, trade, gathering, and building decisions.

In this thesis, multi-agent PPO is implemented through RLlib’s `PPOTrainer` interface [42]. This corresponds to the older RLlib Trainer API used by the original AI Economist-style implementation; newer versions of RLlib organize PPO under the `ray.rllib.algorithms.ppo` API.

A.7 Action Space Specification

The implemented simulations use multi-discrete action spaces for both mobile agents and regional planners. In a multi-discrete action space, the policy selects one option from each action component at every timestep. Each component has its own no-op option, so the agent or planner can leave that component inactive while still selecting actions for other components.

For mobile agents, the action space consists of the following components:

Action component	Number of choices
Build	2
Buy Stone	12
Sell Stone	12
Buy Wood	12
Sell Wood	12
Gather	5
CrossWaterTravel	2
Total action logits	57

The buy and sell components have 12 choices because they include the 11 possible price points from 0 to 10, plus a no-op option. Similarly, the build and travel components each include an inactive/no-op choice. Thus, the mobile-agent policy outputs 57 action logits in total.

For planners, the action-mask/logit size is 308, corresponding to

$$14 \times 22 = 308.$$

Conceptually, each planner chooses marginal tax rates for seven income brackets. However, in the implementation the planner action space contains bracket-action keys from both regional tax components. Each bracket component has 22 choices: one no-op option plus 21 discrete tax-rate choices,

$$\{0.00, 0.05, 0.10, \dots, 1.00\}.$$

Action masks ensure that each planner can only apply the relevant valid tax-bracket actions at valid tax-decision times. At other timesteps, the valid planner action is effectively a no-op.

The spatial observations differ between mobile agents and planners. Mobile agents observe a local spatial window, while planners observe a regional map.

For mobile agents, the spatial observation is:

$$\text{agent world-map} : 7 \times 11 \times 11,$$

consisting of six environment/entity channels plus one padding or boundary channel used near map edges. The corresponding index map is:

$$\text{agent world-idx-map} : 2 \times 11 \times 11,$$

where the two index channels represent:

1. owner map, indicating ownership of private landmarks such as houses,
2. location map, indicating where agents are located.

For planners, the spatial observation is:

$$\text{planner world-map} : 6 \times 25 \times 25,$$

with the following channels:

1. Stone
2. Wood
3. House
4. Water
5. StoneSourceBlock
6. WoodSourceBlock

The planner index map is:

$$\text{planner world-idx-map} : 2 \times 25 \times 25,$$

with the same two index channels:

1. owner map, indicating ownership of private landmarks in the form of houses,
2. location map, indicating where agents are located.

Thus, mobile agents observe a local 11×11 view, while planners observe a regional 25×25 view. The mobile-agent map has seven map channels and two index channels, whereas the planner map has six map channels and two index channels.

Each planner also has fixed planner-side per-agent observation slots, denoted $p0, \dots, p7$. These slots exist for all eight mobile agents. For agents currently outside the planner's region, the corresponding slot is zero-masked. For agents currently inside the planner's region, the slot contains that agent's planner-side information. This keeps the planner observation structure fixed while ensuring that planner-side per-agent information is region-specific.

A.8 Binary Logistic Regression

The regression analysis is conducted at the agent-period level. The dependent variable is

$$Y_{i,p,r} = \begin{cases} 1 & \text{if agent } i \text{ travels at least once in tax period } p \text{ in rollout } r, \\ 0 & \text{otherwise.} \end{cases}$$

In the current simulations, the travel cooldown is approximately one tax period, so an agent can travel at most once within a tax period. More generally, however, the variable should be interpreted as an indicator for whether any travel occurs during the period.

The main regression specification includes the following predictors:

$$X_1 = \text{mean coin},$$

$$X_2 = \text{period coin change},$$

$$X_3 = \text{other minus current local market price},$$

$$X_4 = \text{mean visible resources},$$

$$X_5 = \text{mean visible other houses},$$

$$X_6 = \text{mean visible own houses},$$

$$X_7 = \text{skill}.$$

The variable period coin change is defined as

$$\Delta x_{i,p}^c = x_{i,\text{end of } p}^c - x_{i,\text{start of } p}^c.$$

It is therefore a net change in coin holdings over the tax period, not pure labor income, build income, or trade income. It reflects both coin earned and coin spent during the period.

The market-price variable is defined as

$$\Delta q_{i,p} = q_{i,p}^{\text{other}} - q_{i,p}^{\text{current}},$$

where $q_{i,p}^{\text{current}}$ is the average within-region market price in the agent's current region during period p , and $q_{i,p}^{\text{other}}$ is the corresponding within-region market price in the other region. A positive value means that the other region has a higher local market price than the agent's current region, while a negative value means that the other region has a lower local market price. If a region-period has no relevant trade observations, the plotting code uses period-level fallback averages where possible and otherwise sets the difference to zero.

Before fitting the model, each continuous predictor is standardized:

$$Z_j = \frac{X_j - \mu_j}{\sigma_j}.$$

Thus, the estimated coefficients describe the association between travel and a one-standard-deviation increase in the corresponding variable.

Without agent fixed effects, the model is

$$P(Y_{i,p,r} = 1) = \text{logistic} \left(\beta_0 + \beta_1 Z_{\text{coin}} + \beta_2 Z_{\text{coin change}} + \beta_3 Z_{\text{market price difference}} + \beta_4 Z_{\text{visible resources}} + \beta_5 Z_{\text{visible other houses}} + \beta_6 Z_{\text{visible own houses}} + \beta_7 Z_{\text{skill}} + \gamma_p \right),$$

where γ_p denotes tax-period fixed effects when these are included.

The logistic function is

$$\text{logistic}(x) = \frac{1}{1 + e^{-x}}.$$

Equivalently, the model is linear in log-odds:

$$\log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \sum_j \beta_j Z_j + \gamma_p.$$

The agent fixed-effects specification adds an agent-specific intercept:

$$P(Y_{i,p,r} = 1) = \text{logistic} \left(\sum_j \beta_j Z_j + \gamma_p + \alpha_i \right),$$

where α_i denotes the fixed effect for agent i . In this specification, skill is omitted because it is constant within each agent and is therefore absorbed by the agent fixed effect.

Agent fixed effects control for stable differences between agents, such as persistent travel tendencies, skill level, starting position, long-run role in the economy, or systematic proximity to the regional border. The fixed-effects model therefore asks a within-agent question: when the same agent faces different conditions across tax periods, which variables are associated with travel?

Tax-period fixed effects control for period-wide differences shared by agents in the same period, such as changes in resource abundance, economy-wide wealth, tax schedules, or differences between early and late stages of an episode.

The model chooses the coefficients β that maximize the logistic likelihood. For observation n , let

$$p_n = P(Y_n = 1).$$

The likelihood contribution is

$$p_n^{Y_n} (1 - p_n)^{1 - Y_n},$$

and the full log-likelihood is

$$\ell(\beta) = \sum_n [Y_n \log(p_n) + (1 - Y_n) \log(1 - p_n)].$$

A coefficient β_j is the change in log-odds associated with a one-standard-deviation increase in predictor j , holding the other included variables fixed. To make this easier to interpret, the plots report odds ratios:

$$\text{odds ratio}_j = e^{\beta_j}.$$

An odds ratio above one indicates a positive association with travel, while an odds ratio below one indicates a negative association.

For example, if

$$\beta_{\text{resources}} = -0.40,$$

then

$$e^{-0.40} \approx 0.67.$$

This means that a one-standard-deviation increase in visible resources is associated with the odds of travel being multiplied by 0.67, holding the other included variables fixed.

For each coefficient, the null hypothesis is

$$H_0 : \beta_j = 0,$$

which is equivalent to

$$H_0 : e^{\beta_j} = 1.$$

A small p-value indicates stronger statistical evidence against the null hypothesis of no conditional association. The dashed reference line in the p-value plots marks

$$p = 0.05.$$

The reported p-values are cluster-robust. In the default specification, clustering is by agent, allowing repeated observations from the same agent across tax periods to be correlated. Cluster-robust p-values adjust the estimated uncertainty around the coefficients, but they do not change the coefficient estimates themselves.

This is distinct from including agent fixed effects. Cluster-robust standard errors adjust inference, while agent fixed effects change the regression specification by adding an agent-specific baseline. Thus, the pooled model with cluster-robust p-values still uses cross-agent variation, whereas the fixed-effects model focuses on within-agent variation over time.

Finally, the regression results should be interpreted as descriptive conditional associations rather than causal effects. The model shows how travel is associated with observed variables, holding the included controls fixed, but it does not identify the causal effect of changing those variables.

A.8.1 Fixed Effects: explanation

Tax-period fixed effects give each tax period its own intercept. Without tax-period fixed effects, the regression has one common baseline travel probability:

$$\text{logit}(P(Y = 1)) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

With tax-period fixed effects, it becomes:

$$\text{logit}(P(Y_{i,p,r} = 1)) = \beta_1 X_{1,i,p,r} + \cdots + \beta_k X_{k,i,p,r} + \gamma_p$$

where γ_p is a separate baseline effect for tax period p . This controls for things that affect all agents in the same period, such as: the economy becoming richer over time, resources becoming more or less available, agents being more likely to travel early or late, tax policies changing over the episode, market conditions changing over time. So tax-period fixed effects ask: Within the same tax period, which variables are associated with travel? Agent fixed effects give each agent its own baseline tendency to travel. With agent fixed effects, the model becomes:

$$\text{logit}(P(Y_{i,p,r} = 1)) = \beta_1 X_{1,i,p,r} + \cdots + \beta_k X_{k,i,p,r} + \gamma_p + \alpha_i$$

where α_i is a separate baseline effect for agent i . This controls for stable differences between agents, such as: skill level, typical location, persistent role in the economy, long-run wealth differences, learned behavioral tendencies, being naturally closer to the border. So agent fixed effects ask: When the same agent experiences different conditions over time, which variables are associated with travel? This is why skill drops out when agent fixed effects are included: skill is constant for each agent, so it is already captured by the agent's fixed effect. In Simple Terms: Tax-period fixed effects compare agents within the same period. Agent fixed effects compare an agent to itself over time. Together, they make the regression stricter: Is a variable associated with travel when comparing the same agent across different periods, while also accounting for period-wide changes in the economy? That is why the fixed-effects regression is often more cautious than the pooled regression.

A.8.2 Estimation Procedure for the Logistic Regression Models

The regression models used in the analysis are binary logistic regressions. The dependent variable is an indicator for whether agent i travels at least once during tax period p in rollout r :

$$Y_{i,p,r} = \begin{cases} 1 & \text{if agent } i \text{ travels during tax period } p \text{ in rollout } r, \\ 0 & \text{otherwise.} \end{cases}$$

Because the outcome variable is binary, the model is estimated using maximum likelihood estimation rather than ordinary least squares. The logistic model assumes that the conditional probability of travel is given by

$$P(Y_{i,p,r} = 1 \mid \mathbf{Z}_{i,p,r}) = \Lambda(\eta_{i,p,r}),$$

where

$$\Lambda(x) = \frac{1}{1 + \exp(-x)}$$

is the logistic function, and $\eta_{i,p,r}$ is the linear predictor.

For the pooled specification, the linear predictor is

$$\eta_{i,p,r} = \beta_0 + \mathbf{Z}'_{i,p,r}\boldsymbol{\beta} + \gamma_p,$$

where $\mathbf{Z}_{i,p,r}$ is the vector of standardized covariates, $\boldsymbol{\beta}$ is the corresponding coefficient vector, and γ_p denotes tax-period fixed effects when included. The standardized covariates include mean coin, period coin change, visible resources, visible houses, and the regional market-price difference. In specifications without agent fixed effects, building skill is also included.

For the agent fixed-effects specification, the linear predictor is

$$\eta_{i,p,r} = \mathbf{Z}'_{i,p,r}\boldsymbol{\beta} + \gamma_p + \alpha_i,$$

where α_i is an agent-specific fixed effect. The fixed effect gives each agent its own baseline tendency to travel, thereby controlling for stable differences between agents such as skill, starting position, persistent behavioral tendencies, or long-run differences in economic role. Since skill is constant within each agent, it is absorbed by α_i and cannot be separately estimated in the fixed-effects specification.

The likelihood contribution of each observation is

$$P(Y_{i,p,r} = y_{i,p,r}) = [\Lambda(\eta_{i,p,r})]^{y_{i,p,r}} [1 - \Lambda(\eta_{i,p,r})]^{1-y_{i,p,r}}.$$

The full likelihood is the product of these terms across all observations:

$$L(\boldsymbol{\beta}) = \prod_{i,p,r} [\Lambda(\eta_{i,p,r})]^{Y_{i,p,r}} [1 - \Lambda(\eta_{i,p,r})]^{1-Y_{i,p,r}}.$$

Equivalently, the log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i,p,r} [Y_{i,p,r} \log \Lambda(\eta_{i,p,r}) + (1 - Y_{i,p,r}) \log(1 - \Lambda(\eta_{i,p,r}))].$$

The coefficient estimates are chosen to maximize this log-likelihood:

$$\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta}).$$

Thus, the estimator selects the parameters that make the observed pattern of travel and non-travel outcomes most likely under the logistic model.

The reported coefficients are transformed into odds ratios:

$$\text{odds ratio}_j = \exp(\hat{\beta}_j).$$

An odds ratio above one indicates that a one-standard-deviation increase in the corresponding covariate is associated with higher odds of travel, holding the other included variables fixed. An odds ratio below one indicates lower odds of travel.

Inference is based on cluster-robust standard errors. In the default specification, observations are clustered by agent, allowing repeated observations from the same agent across tax periods to be correlated. The cluster-robust correction changes the estimated uncertainty and p-values, but not the coefficient estimates themselves.

The resulting estimates should be interpreted as descriptive conditional associations rather than causal effects. The logistic model identifies how travel decisions are statistically associated with observed variables in the simulated data, conditional on the included controls and fixed effects. It does not identify the causal effect of changing those variables, since the regressors are themselves generated endogenously by the learned policies and the evolving simulated economy.

A.9 Shared Learning Parameters

Parameter sharing in mobile agents improves learning efficiency because all agents can learn from each other’s experiences. The main reason this is feasible, is because agents operate under identical rules and differ primarily in their states, locations, and skill levels. In this setting, a shared policy allows experience collected from all agents to contribute to a single learning process, significantly accelerating convergence compared to learning independent policies.

Beyond computational benefits, parameter sharing also serves as an inductive bias: it encourages agents to learn general behavioral patterns that are robust across different economic conditions. Because agents are exposed to diverse situations—such as varying resource availability, market states, and tax regimes—the shared policy learns to condition decisions on observations rather than relying on fixed, agent-specific behaviors. This is particularly important in a dynamic environment with mobility and cross-regional interaction, where agents may encounter a wide range of economic contexts over time that differ from their original learning environment.

However, this design requires the shared policy to represent multiple behavioral modes simultaneously, such as specialization in production or resource gathering. When agents face different constraints or roles, this can introduce gradient interference, as a single policy must approximate distinct strategies within one parameterization. Despite this limitation, the benefits of improved generalization, stability, and sample efficiency make parameter sharing well-suited for the present setting.

A.10 Parallel Training Configuration and Parameter Interpretation

This section explains the key training parameters used in the implementation and how they relate to data collection and PPO optimization. The values described here correspond to the current non-local `simulation_run.py` configuration.

A.10.1 Workers

The parameter `num_workers` specifies how many parallel rollout worker processes are used during training. Each worker runs an independent copy of the environment and collects experience by interacting with it using the current policies.

In the current non-local configuration,

`num_workers = 4.`

Increasing the number of workers increases the amount of experience that can be collected in parallel, but also increases CPU and memory requirements.

A.10.2 Environments per Worker

The parameter `num_envs_per_worker` determines how many environments are executed within each worker process. In the current configuration,

$$\text{num_envs_per_worker} = 1.$$

The total number of parallel environment replicas is therefore

$$\text{total environments} = \text{num_workers} \times \text{num_envs_per_worker} = 4 \times 1 = 4.$$

A.10.3 Rollout Fragment Length

The parameter `rollout_fragment_length` specifies how many timesteps each environment executes before returning collected experience to the trainer. Rather than waiting for a full episode to complete, RLlib collects shorter rollout fragments.

In the current configuration,

$$\text{rollout_fragment_length} = 200.$$

Thus, each environment contributes rollout fragments of approximately 200 timesteps. Since the episode length is 1000, one complete episode corresponds to approximately

$$\frac{1000}{200} = 5$$

rollout fragments per environment.

A.10.4 Training Batch Size

The parameter `train_batch_size` defines how many transitions are collected before one PPO training iteration is performed. In the current configuration,

$$\text{train_batch_size} = 800.$$

This is consistent with the rollout setup:

$$\text{train_batch_size} = \text{num_workers} \times \text{num_envs_per_worker} \times \text{rollout_fragment_length} = 4 \times 1 \times 200 = 800.$$

Thus, each PPO iteration uses approximately 800 newly collected transitions.

A.10.5 Mini-batch Size

The parameter `sgd_minibatch_size` determines how the training batch is split during optimization. Instead of updating the model once using the full training batch, PPO divides the batch into smaller mini-batches.

In the current configuration,

$$\text{sgd_minibatch_size} = 200.$$

Given a training batch size of 800, this gives

$$\frac{800}{200} = 4$$

mini-batches per pass through the training batch.

A.10.6 Number of SGD Iterations

The parameter `num_sgd_iter` specifies how many times PPO reuses the same collected batch for optimization. In the current configuration,

$$\text{num_sgd_iter} = 4.$$

The total number of mini-batch gradient updates per PPO training iteration is therefore

$$\frac{\text{train_batch_size}}{\text{sgd_minibatch_size}} \times \text{num_sgd_iter} = \frac{800}{200} \times 4 = 16.$$

This reuse of rollout data improves sample efficiency by extracting more learning signal from each collected batch.

A.10.7 Relation to Episode Length and Tax Period

The episode length is

$$1000$$

timesteps, while the tax period is

$$100$$

timesteps. Because the rollout fragment length is 200, each rollout fragment contains approximately two tax periods. A full episode contains ten tax periods and corresponds to approximately five PPO rollout fragments per environment.

A.10.8 Relation to the Original Training Procedure

The original AI Economist training setup also uses multiple parallel replicas of the environment. In the current implementation, the number of replicas is given by

$$\text{replicas} = \text{num_workers} \times \text{num_envs_per_worker}.$$

The number of steps collected per replica corresponds to `rollout_fragment_length`, and the total number of transitions collected before an update corresponds to `train_batch_size`. Mini-batch size and repeated optimization over the same rollout data are controlled by `sgd_minibatch_size` and `num_sgd_iter`. Thus, although the exact numerical values differ from the original AI Economist configuration, the overall structure remains the same: collect rollout data from parallel environments, concatenate it into a training batch, and perform multiple PPO gradient updates.

A.10.9 Summary

The training system consists of two main stages: parallel data collection and batched PPO optimization. The worker and environment parameters determine how much data is collected in parallel, while the batch size, mini-batch size, and number of SGD passes determine how that data is reused for learning.